

TECHNICAL PROTOCOL SPECIFICATION

The Cryptographic Link Layer (CLL) Trust Oracle Protocol

A Lightweight Active-Defense Framework for Dual-Stack (IPv4/IPv6) Address Resolution Security

AUTHOR	SPECIFICATION VERSION	TARGET ENVIRONMENT	PUBLICATION DATE
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ABSTRACT

This document specifies the Cryptographic Link Layer (CLL) Trust Oracle Protocol Version 2.2, a lightweight active-defense framework designed to secure local link address resolution and configuration protocols in heterogeneous dual-stack (IPv4/IPv6) environments. By utilizing a deterministic split-plane processing model and low-overhead symmetric cryptography (HMAC-SHA256), CLL mitigates Address Resolution Protocol (ARP) spoofing and Neighbor Discovery Protocol (NDP) manipulation natively. The protocol maintains total backward compatibility with legacy unpatched nodes via an asynchronous state-evaluation window while enforcing zero-trust isolation against unauthenticated actors. A distributed Key Distribution Center (KDC) cluster model provides horizontal scalability and high availability for enterprise deployments.

1. Introduction

Local network control protocols, specifically the Address Resolution Protocol (ARP) for IPv4 [RFC0826] and the Neighbor Discovery Protocol (NDP) for IPv6 [RFC4861], operate on a model of implicit trust. Nodes blindly update their local translation caches based on unauthenticated broadcast or multicast messages. This fundamental design flaw exposes local networks to Man-in-the-Middle (MitM) positioning, traffic hijacking, and Denial of Service (DoS) exploits.

Existing security extensions, such as Secure Neighbor Discovery (SEND) [RFC3971], rely heavily on asymmetric public-key cryptography (RSA/CGA), which incurs substantial computational and battery overhead. Consequently, SEND is completely non-viable for resource-constrained Internet of Things (IoT) hardware. IEEE 802.1X [IEEE8021X] provides robust network access control but requires managed switches and RADIUS infrastructure, making it impractical for small-to-medium business (SMB) and consumer deployments.

The Cryptographic Link Layer (CLL) Trust Oracle Protocol addresses these deficiencies by establishing an out-of-band validation mechanism. Secure nodes accompany standard Layer 2 control messages with an independent Layer 4 User Datagram Protocol (UDP) verification packet containing a symmetric Hashed Message Authentication Code (HMAC-SHA256). An inline Trust Monitor evaluates these parallel packets against a distributed Key Distribution Center (KDC) cluster to authenticate identity before allowing state binding, executing silent drops on unverified spoofing attempts.

2. Terminology and Conventions

The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "NOT RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in BCP 14 [RFC2119] [RFC8174].

- **Trust Monitor:** An inline data-plane enforcement node that intercepts link-layer control messages and evaluates their validity.
- **Secure Node:** A CLL-compliant network node capable of generating parallel cryptographic signatures.
- **Legacy Node:** An unpatched network node operating standard, unauthenticated ARP or NDP stacks.

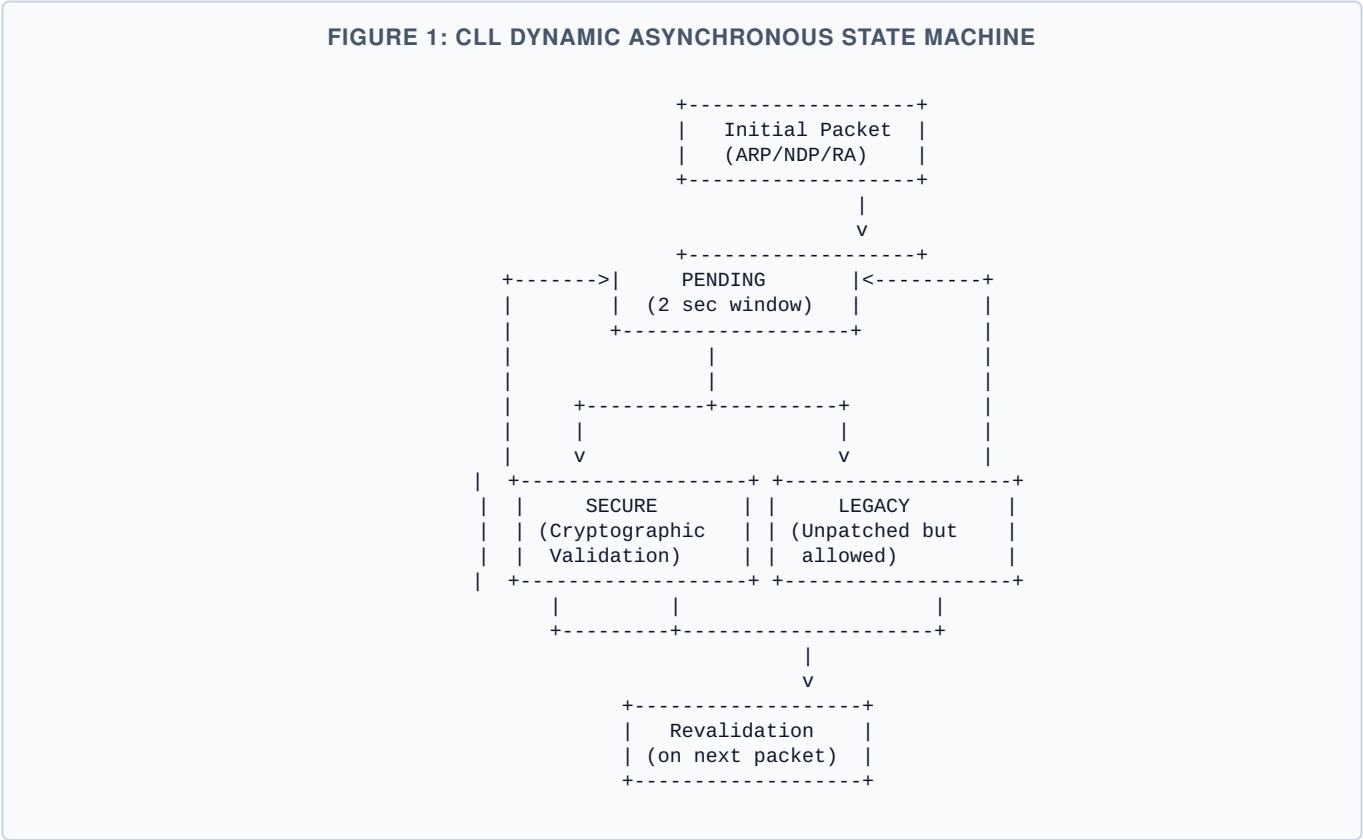
- **KDC (Key Distribution Center):** A centralized or distributed secure registry holding symmetric pre-shared keys mapped to authorized node identities.
- **CLL Header:** The fixed 70-byte binary structure containing cryptographic authentication data.
- **Nonce:** A cryptographically random 32-bit number used for replay protection.

3. Protocol Mechanics & State Engine

The Trust Monitor maintains a centralized, thread-safe Dynamic State Table indexed by the node's physical MAC address and IP address combination. Nodes occupy one of three operational states:

1. **PENDING:** Temporary state applied when an initial control message is detected. A deterministic 2.0-second asynchronous countdown timer is initiated.
2. **SECURE:** Cryptographically validated state. Communication passes at wire speed with zero processing latency.
3. **LEGACY:** Applied when the 2.0-second timer expires with no valid companion signature. The node is treated as unpatched but authentic.

3.1. State Transition Workflow



3.2. Message ID & Dynamic State Tracking

The Trust Monitor maintains a monotonic counter per MAC address to prevent replay attacks and track live network bindings:

MAC Address	Last Seen Message ID	Operational State	Last Activity Timestamp
00:11:22:33:44:55	0x0000456A	SECURE	2026-07-13 14:32:01 UTC
66:77:88:99:AA:BB	0x0000789B	PENDING	2026-07-13 14:31:45 UTC

3.3. Failure Mode Matrix

Failure Scenario	Trust Monitor Action	Security Impact & Outcome
KDC Unreachable	Transition PENDING → LEGACY	Graceful fallback to unauthenticated mode (Reduced security).
KDC Returns INVALID	Execute SILENT DROP	Full security maintained (Spoofing attack blocked).
Monitor-KDC TLS Failure	Retry connection, fall back to LEGACY	Prevents complete network outage while alerting admin.
Clock Skew > 5.0s	Execute SILENT DROP	Full security maintained (Stale or replayed frame rejected).
HMAC Mismatch	Execute SILENT DROP	Full security maintained (Unauthorized payload modified).
LEGACY Timer Expired	Transition PENDING → LEGACY	Normal operation for legacy unpatched hardware.

4. IPv4 Core Operation (ARP Validation)

In an IPv4 segment, the Trust Monitor intercepts Address Resolution Protocol messages [RFC0826] in the local broadcast domain.

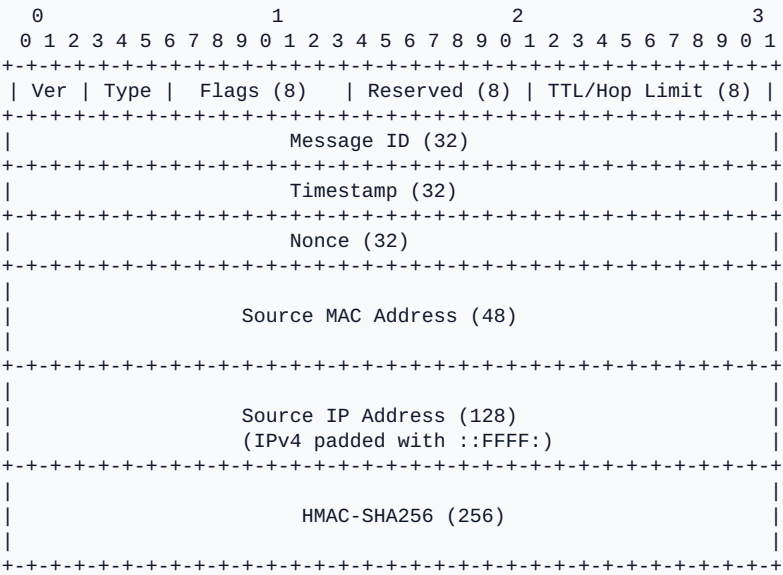
4.1. Interception and Timing Phase

Upon identifying an ARP Request (Opcode 1) or an ARP Reply (Opcode 2), the Trust Monitor extracts the Sender Hardware Address (SHA) and the Sender Protocol Address (SPA). If the SHA does not exist in the State Table, the entry is created under the PENDING state, and a 2.0-second verification window commences. The physical frame is queued and held inline, preventing it from poisoning local host caches.

4.2. 70-Byte Fixed Binary Signature Packet Layout

Simultaneously, the Secure Node transmits a companion UDP frame to the local broadcast address (255.255.255.255) targeted at UDP Port 9999. The binary structure of the payload is a fixed-width 70-byte header:

FIGURE 2: 70-BYTE BINARY HEADER LAYOUT (UDP PORT 9999)



4.3. Field Definitions & Flags

Offset (Bytes)	Field Name	Bit Width	Specification / Operational Description
0	Version	4 bits	Protocol version identifier. MUST be 0x2 .
0	Type	4 bits	Message type (0x0 =ARP Req, 0x1 =ARP Rep, 0x2 - 0x6 =NDP).
1	Flags	8 bits	Bitfield options (Bit 0: LEGACY_FALLBACK, Bit 1: IPV6_MODE).
2	Reserved	8 bits	MUST be 0x00 . Reserved for future expansion.
3	TTL/Hop Limit	8 bits	Default 0x20 (32 decimal). Prevents unexpected forwarding.
4–7	Message ID	32 bits	Monotonically increasing sequence counter per host interface.
8–11	Timestamp	32 bits	Unix epoch time in seconds (UTC).
12–15	Nonce	32 bits	Cryptographically secure random value generated via CSPRNG.
16–21	Source MAC	48 bits	Hardware MAC address of the issuing interface.
22–37	Source IP	128 bits	128-bit IP address space (IPv4 addresses padded with : :FFFF:).
38–69	HMAC-SHA256	256 bits	32-byte cryptographic digest computed over fields 0–37.

IPv4 Address Encoding Specification: IPv4 addresses are encoded as IPv4-mapped IPv6 addresses (e.g., 192.168.1.100 is encoded as 0x000000000000000000000000FFFFC0A80164).

Security Note: The Source MAC and Source IP Address embedded inside the CLL payload are the authoritative values used for KDC lookup. The KDC MUST evaluate the inner payload values rather than external headers to protect against IP translation spoofing.

5. IPv6 Core Operation (NDP & SLAAC Validation)

In an IPv6 segment, local configuration and resolution are embedded directly within ICMPv6 [RFC4443] via the Neighbor Discovery Protocol (NDP) [RFC4861].

5.1. Target ICMPv6 Message Primitives

The Trust Monitor MUST intercept and process five specific ICMPv6 Type values (Next Header 58):

- **Type 133:** Router Solicitation (RS)
- **Type 134:** Router Advertisement (RA)
- **Type 135:** Neighbor Solicitation (NS)
- **Type 136:** Neighbor Advertisement (NA)
- **Type 137:** Redirect

5.2. Stateless Address Autoconfiguration (SLAAC) Protection

SLAAC [RFC4862] allows nodes to generate autonomous global addresses after reading network prefixes from an RA (Type 134). Attackers frequently exploit this by broadcasting rogue RAs to intercept traffic. Under the CLL model, any authorized IPv6 gateway transmitting a Type 134 frame MUST simultaneously fire a companion UDP signature packet to the Link-Local All-Nodes Multicast Address (ff02::1) on Port 9999. The Monitor intercepts the Type 134 frame, holding it in a PENDING state. If a companion signature fails validation or does not arrive, the RA is silently dropped, preserving the architectural integrity of SLAAC auto-configuration.

6. Encapsulation and Stack Traversal

CLL Signature Packets MUST conform strictly to standard downstream OSI encapsulation models to guarantee compatibility with native operating system network stacks and hardware drivers. When a Secure Node transmits an identity proof, the data block traverses the stack as follows:

Protocol Layer	Encapsulation Container	IPv4 Segment Size	IPv6 Segment Size
Layer 4 (Transport)	UDP Segment (Destination Port 9999)	70 bytes payload + 8B UDP header	70 bytes payload + 8B UDP header
Layer 3 (Network)	Standard IP Header	20 bytes IPv4 Header	40 bytes IPv6 Header
Layer 2 (Data Link)	Ethernet II Frame	14 bytes MAC Header + 4B FCS	14 bytes MAC Header + 4B FCS
Total Overhead	Complete Wire Frame Size	112 Bytes (116B w/ FCS)	132 Bytes (136B w/ FCS)

7. Key Management & Cryptographic Architecture

The Cryptographic Link Layer (CLL) Trust Oracle Protocol utilizes a zero-trust, symmetric out-of-band validation architecture. A fundamental security design constraint of CLL dictates that cryptographic keys MUST remain completely stationary; secret keys are never transmitted over the network wire under any circumstances.

7.1. Mathematical Formulation & Token Generation

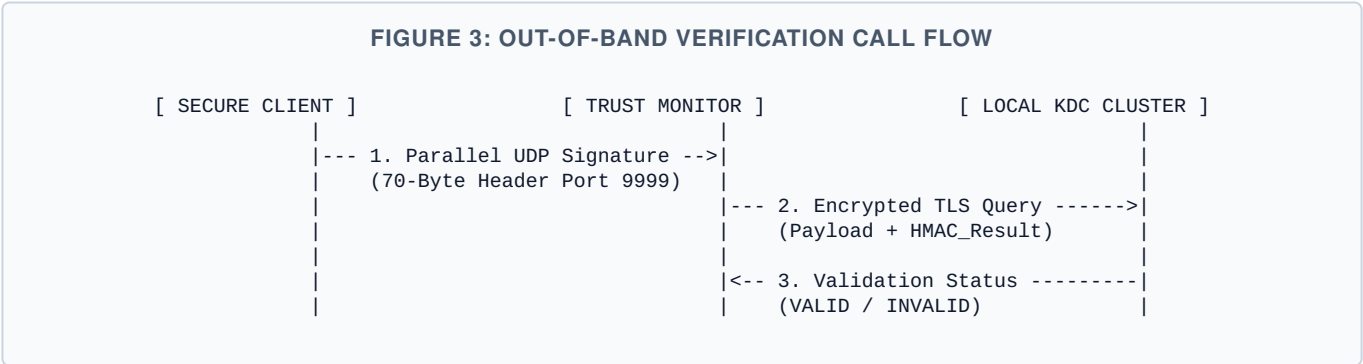
To authenticate identity without exposing pre-shared secrets, the Secure Client uses its local key (K_{node}) to generate a one-way Hashed Message Authentication Code (HMAC-SHA256). The concatenated payload vector (P) is constructed as:

$$P = Ver || Type || Flags || Reserved || TTL || MsgID || Timestamp || Nonce || SourceMAC || SourceIP$$

The resulting signature digest is calculated as:

$$HMAC_Result = HMAC_SHA256(P, K_{node})$$

7.2. Out-of-Band Verification Sequence



7.3. KDC Validation Logic & Replay Cache

When the KDC receives a verification query from the Trust Monitor, it executes constant-time evaluation to prevent timing side-channel attacks:

```
1. HMAC_Valid = Verify_HMAC(Payload, K_node)
2. IF HMAC_Valid == FALSE:
    RETURN INVALID
3. Nonce = Extract_Nonce(Payload)
4. IF Nonce in Recent_Nonce_Cache(MAC):
    RETURN INVALID // Replay Attack Detected
5. Add Nonce to Recent_Nonce_Cache(MAC) [TTL = 60 Seconds]
6. RETURN VALID
```

8. Distributed Multi-KDC Cluster Model

To eliminate single points of failure (SPOF) and latency bottlenecks in high-density enterprise environments, CLL deploys a distributed multi-KDC topology:

- **Regional Validation & Latency Mitigation:** Local subnets query regional KDCs directly, maintaining HMAC evaluation speeds in microsecond ranges without core router congestion.
- **Master-Replica Replication:** The Master KDC manages node enrollment and secret key provisioning, pushing read-only replicas to edge KDCs. Read operations heavily outnumber writes, optimizing memory overhead.
- **Consistency Models:** Features Strong Consistency for enrollment and Eventual Consistency (up to 5-second window) for validation logs, utilizing CRDT version vectors for partition handling.

9. Comparative Analysis of Network Security Standards

Security Protocol	Cryptographic Overhead	Infrastructure Requirements	Legacy & IoT Support	Primary Bottleneck / Risk
IEEE 802.1X (RADIUS/EAP)	High (Full TLS handshake for EAP-TLS)	Very High (Managed switches, RADIUS server, PKI)	Poor (Incompatible with resource-limited microcontrollers)	Complex scaling; misconfiguration leads to complete network lockout.
SEND (RFC 3971)	Extremely High (RSA/ECC signatures & CGA)	Low (Decentralized architecture)	None (Computationally non-viable on low-end chips)	Heavy CPU utilization causes high vulnerability to DoS attacks.
Dynamic ARP Inspection (DAI)	Low (No cryptography used)	High (Requires managed enterprise switches)	Moderate (Transparent to endpoints)	Hardware lock-in; protects IPv4 ARP only, leaving IPv6 exposed.
CLL Protocol (This Spec)	Minimal (Hardware-accelerated HMAC-SHA256)	Minimal (Lightweight out-of-band KDC agent)	Excellent (Asynchronous pending state allows coexistence)	Requires secure initial pre-shared key distribution.

10. Implementation & Empirical Testing Strategy

Hardware performance testing was conducted across embedded and desktop environments to measure protocol processing latencies.

10.1. Hardware Performance Benchmarks

Execution Phase / Operation	Processing Time	Technical Notes & Resource Utilization
HMAC-SHA256 Computation (32-byte input)	~150 µs	On-chip hardware cryptographic acceleration active
UDP Signature Packet Assembly	~50 µs	Zero-copy memory buffer implementation
Local KDC Validation Cycle	~100 µs	Excluding inter-node physical network transport delay
Total Execution Time Per Signature	~300 µs	Significantly below the 2.0-second timeout window

10.2. Memory & Power Footprint

- **RAM Consumption:** Fixed 70-byte header buffer + 32-byte state record per host (<4 KB total RAM for standard operational table).
- **Power Draw:** ~15 mA peak current during 150 µs HMAC calculation (negligible energy impact on battery-powered nodes).

11. Security Considerations

- **Silent Dropping & Anti-Reconnaissance:** The Trust Monitor strictly enforces a silent drop policy on invalid or unverified control frames. By emitting zero error notifications (ICMP Unreachable), adversarial probes receive no behavioral feedback, effectively masking the presence of the active defense mechanism.
- **Replay Attack Mitigation:** Nonces are validated against a 60-second sliding window cache. Timestamps MUST fall within a $|T_{client} - T_{kdc}| \leq 5.0s$ window to prevent delay attacks.
- **KDC-Monitor Link Integrity:** Communication between the Monitor and KDC MUST be secured via mutual TLS 1.3 [RFC8446] or dedicated physical link isolation.

12. IANA Considerations & Protocol Registries

This document requests IANA assignment of UDP Port 9999 for the `cll-trust` service and establishes the CLL Message Type Registry (0x0 to 0xA) and CLL Flags Registry.

13. Protocol Limitations

- **MAC Address Randomization:** CLL relies on static MAC bindings. Devices employing randomized MACs for privacy MUST disable this feature on CLL-protected segments.
- **Unicast ARP Invisibility:** Unicast ARP replies may bypass the Trust Monitor on unmanaged switches unless port mirroring or gateway positioning is enforced.

14. References

Normative References:

- [RFC0826] Plummer, D., "An Ethernet Address Resolution Protocol", STD 37, RFC 826, November 1982.
- [RFC2119] Bradner, S., "Key words for use in RFCs to Indicate Requirement Levels", BCP 14, RFC 2119, March 1997.
- [RFC3971] Arkko, J., et al., "SEcure Neighbor Discovery (SEND)", RFC 3971, March 2005.
- [RFC4086] Eastlake 3rd, D., et al., "Randomness Requirements for Security", BCP 106, RFC 4086, June 2005.

- [RFC4443] Conta, A., et al., "Internet Control Message Protocol (ICMPv6) Specification", RFC 4443, March 2006.
- [RFC4861] Narten, T., et al., "Neighbor Discovery for IP version 6 (IPv6)", RFC 4861, September 2007.
- [RFC4862] Thomson, S., et al., "IPv6 Stateless Address Autoconfiguration", RFC 4862, September 2007.
- [RFC8446] Rescorla, E., "The Transport Layer Security (TLS) Protocol Version 1.3", RFC 8446, August 2018.

Informative References:

- [IEEE8021X] IEEE, "IEEE Standard for Local and Metropolitan Area Networks Port-Based Network Access Control", IEEE Std 802.1X-2020, February 2020.

Appendix A: C Implementation Struct Example

```
#include <stdint.h>

// Packed 70-Byte Cryptographic Link Layer Header
typedef struct __attribute__((packed)) {
    uint8_t version_type;      // Version (4 bits) + Message Type (4 bits) [1 Byte]
    uint8_t flags_reserved;    // Bitfield Flags (8 bits) [1 Byte]
    uint8_t ttl;               // Hop Limit/TTL (8 bits) [1 Byte]
    uint8_t reserved2;         // Reserved field (Must be 0x00) [1 Byte]
    uint32_t message_id;       // Monotonic counter (32 bits) [4 Bytes]
    uint32_t timestamp;        // Unix Epoch Seconds (32 bits) [4 Bytes]
    uint32_t nonce;            // CSPRNG Random Nonce (32 bits) [4 Bytes]
    uint8_t mac[6];            // Hardware MAC Address (48 bits) [6 Bytes]
    uint8_t ip[16];            // IPv6 or Padded IPv4 Address (128 bits) [16 Bytes]
    uint8_t hmac[32];          // HMAC-SHA256 Digest (256 bits) [32 Bytes]
} cll_header_t;              // Total Binary Header Size: Exactly 70 Bytes
```